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Key Points:

- Fractures with mixed-mode displacements of opening and shearing complicate $D_{\max}/L$ scaling analysis
- Field observations indicate the Koa'e Fault Zone forms from the downward propagation of joints into faults at depth
- The $D_{\max}/L$ scaling relationships for all structure types, including joints and faults, show a near-linear power-law relation

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

H. C. J. Cheng,
hhansen2@ua.edu

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Author Contributions:

Conceptualization: Hiu Ching Jupiter Cheng, Christian Klimczak
Data curation: Hiu Ching Jupiter Cheng, Jay Mrazek
Formal analysis: Hiu Ching Jupiter Cheng, Jay Mrazek
Funding acquisition: Hiu Ching Jupiter Cheng
Investigation: Hiu Ching Jupiter Cheng, Jay Mrazek, Christian Klimczak
Methodology: Hiu Ching Jupiter Cheng, Jay Mrazek, Christian Klimczak
Supervision: Christian Klimczak
Visualization: Hiu Ching Jupiter Cheng
Writing – original draft: Hiu Ching Jupiter Cheng

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The Transition From Jointing to Faulting Observed at the Koa'e Fault Zone, Hawai'i Volcanoes National Park, Hawaii

Hiu Ching Jupiter Cheng^{1,2} , Jay Mrazek¹, and Christian Klimczak¹ 

¹Center for Planetary Tectonics at UGA, Department of Geology, University of Georgia, Athens, GA, USA, ²Department of Geological Sciences, University of Alabama, Tuscaloosa, AL, USA

Abstract Fractures can exhibit mixed modes of displacement, that is, combinations of displacements parallel and perpendicular to the fracture plane, that make a displacement-to-length ($D_{\max}/L$) scaling analysis challenging. However, such analysis is important for understanding the propagation and nature of mixed-mode fracture populations. In this study, we investigate the $D_{\max}/L$ scaling relationship for fractures involving opening and shearing modes from field measurements at the spectacularly exposed Koa'e Fault Zone in Hawai'i Volcanoes National Park, Hawaii. Its major structures have prominent fault scarps and display openings of up to several meters. They are locally accompanied by monoclines and sheared and pure joints. Through structural mapping and detailed field observations, we identify a morphological continuum along the structures representing different stages in the evolution of the faults. Contrary to previous studies, our observations support that faults are formed by the downward propagation of joints that transition to faulting at depth, then creating the monoclinical flexure. Our measurements allow us to investigate the $D_{\max}/L$ scaling behavior for the total mixed-mode displacement and their individual vector components, that is, reliefs and openings. The $D_{\max}/L$ scaling relationships for all structure types, including pure joints, sheared joints, and faults, show a power-law relation with a near-linear dependence of maximum displacement and length. The joint apertures scale to length with a nearly linear scaling relationship, not following the widely observed square root scaling relationship that all opening-mode fractures are believed to have.

Plain Language Summary Fractures in rocks often form in complex ways, with displacements both parallel and perpendicular to the fracture plane. This makes it challenging to study how displacement relates to fracture length, but understanding this relationship is crucial for learning how these fractures form and evolve. In this study, we focused on the Koa'e Fault Zone in Hawaii, where fractures vary in length and show large openings and reliefs (elevation difference across fractures). We mapped and measured these structures, identifying different stages of the fault zone evolution. Contrary to previous research, our findings suggest that these fractures begin as surface openings that grow downward and eventually transition into faults that experience sliding. When we analyzed how the displacements of these fractures scale with their length, we found that the opening of the fractures here does not follow the widely observed square root scaling rule but instead scales linearly with length. These findings improve our understanding of how mixed-mode fractures develop and propagate in the Earth's crust, helping us better estimate how fluids move through or are stored in fractured crystalline rock.

1. Introduction

1.1. Displacement Modes of Fractures

Rock fractures are typically categorized based on their different modes of displacement, that is, opening (mode-I), sliding and tearing (modes II and III), and closing modes (mode-IV or anti-mode-I). Opening-mode displacement occurs when the fracture surfaces move away from one another without wall-parallel movement, resulting in an opening referred to as the aperture. Opening-mode fractures include pure joints, veins, and dikes (e.g., Pollard et al., 1982; Pollard & Aydin, 1984; Rubin, 1993; Segall & Pollard, 1983). Sliding- and tearing-mode displacements involve shearing, or parallel movement, between fracture surfaces. Examples of mode II/III fractures include normal, thrust, or strike-slip faults (e.g., Chinnery, 1961, 1965; Davison, 1994; Gupta & Scholz, 2000; Petit & Barquins, 1988; Rodgers, 1980). Closing-mode displacement occurs when the fracture surfaces move toward one another, resulting in porosity loss or dissolution along the fracture plane. Closing-mode fractures include compaction bands and stylolites (e.g., Fletcher & Pollard, 1981; Fueten & Robin, 1992; Mollema & Antonellini, 1996; Zhou & Aydin, 2012; Zhou et al., 2010).

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Christian Klimczak

Although many fractures show displacements that are dominated by one fracturing mode, mixed-mode fractures are also commonly observed in nature. Fractures, such as sheared joints, can display both opening-, and sliding modes (e.g., Engelder, 1999; Pollard et al., 1982), while fractures showing both closing- and sliding modes are frequently observed in porous sandstones as deformation bands (Fossen et al., 2007; Fossen & Hesthammer, 1997; Schultz & Soliva, 2012). Normal faults may also have a dilational component, which may be due to depth-dependent changes in rheology and fault dip along the structure (e.g., Ferrill & Morris, 2003). Dilation associated with normal faulting also occurs when loading conditions become favorable for vertical, downward-propagating joints to be reactivated as normal faults (Gudmundsson, 1992, 2011), a scenario commonly associated with large rifts. Prominent examples include the Almannagjá normal fault at the Reykjanesryggur-Langjökull rift system of southwest Iceland (e.g., Gudmundsson, 1992, 2011), the Koa'e Fault Zone associated with the rift zone at the Kilauea volcano, Hawaii (Denlinger & Okubo, 1995; Ge et al., 2019; Holland et al., 2006; Martel & Langley, 2006), or the Wonjwe Fault Belt in Ethiopia at the East African Rift (e.g., Acocella et al., 2003). Pit crater chains aligned within graben on Mars (Ferrill et al., 2011; Wyrick et al., 2004) and troughs on asteroid 4 Vesta (Cheng & Klimczak, 2022) are also interpreted to display this type of mixed-mode structure.

1.2. Displacement-Length Scaling

Displacements of all modes can be analyzed in terms of the relationship with their lengths, referred to as (maximum) displacement-length ($D_{\max}/L$) scaling. For several fractures within a fracture population, fracture lengths (L), as measured by the length of their tip-to-tip map or outcrop traces, are correlated to the measured maximum displacements ($D_{\max}$). The method is widely used to characterize the growth of fractures. It is applied to learn about mechanical properties and fracture geometries on Earth (e.g., Dawers et al., 1993; Poulimeanos, 2000; Wilkins & Gross, 2002) but also to estimate strain and infer about lithospheric structure on other rocky planets (e.g., Byrne et al., 2021; Polit et al., 2009; Schultz, 1997; Schultz et al., 2006).

Numerous studies have investigated the $D_{\max}/L$ scaling of faults and opening-mode fractures (e.g., Cowie & Scholz, 1992; Delaney & Pollard, 1981; Muraoka & Kamata, 1983; Olson, 2003; Schultz et al., 2006, 2008, 2013; Vermilye & Scholz, 1995). It is widely accepted that within a population of fractures, the maximum displacement is systematically related to length by a power law. For faults, wide consensus exists that $D_{\max}$ scales linearly with L , displaying a power-law exponent of 1 (Figure 1a; e.g., Cowie & Scholz, 1992; Clark & Cox, 1996; Schultz & Fossen, 2002; Schultz et al., 2008; Scholz et al., 1993; Schultz et al., 2013):

$$D_{\max} = \gamma L, \quad (1)$$

where γ is a fault population-specific proportionality coefficient that is related to the mechanical properties of the host rock, geometry of the fault plane, and stress conditions around the fault tips (e.g., Schultz & Fossen, 2002). Linear scaling implies that fault growth is self-similar to smaller and larger faults in the same population sharing the same overall geometry.

For opening-mode fractures, $D_{\max}$ and L are found to show a sublinear scaling relationship with a power-law exponent of 0.5 (Figure 1b; Olson, 2003; Schultz et al., 2013):

$$D_{\max} = \alpha L^{0.5}, \quad (2)$$

where α is the fracture population-specific proportionality coefficient related to the mechanical and fracturing properties of the host rock (Olson, 2003; Schultz et al., 2008). Due to post-jointing relaxation, fracture network geometry, and mechanical interactions among fractures, the scaling of joints but also veins and dikes may display a large scatter, and exponents may be greater than 0.5 (e.g., Olson, 2003; Schultz et al., 2008; Klimczak et al., 2010; also see Figure 1b).

The scaling behavior of mixed-mode fractures is not as extensively studied or well understood as that of opening, or sliding and tearing mode fractures, as accurately measuring or resolving the different components of displacement along these types of geologic structures can be challenging. Schultz et al. (2008) compile displacements and lengths of deformation bands showing that scaling exponents can be 0.5 or 1. Differences are ascribed to the dominant effect of strain components, with linear scaling ascribed to deformation

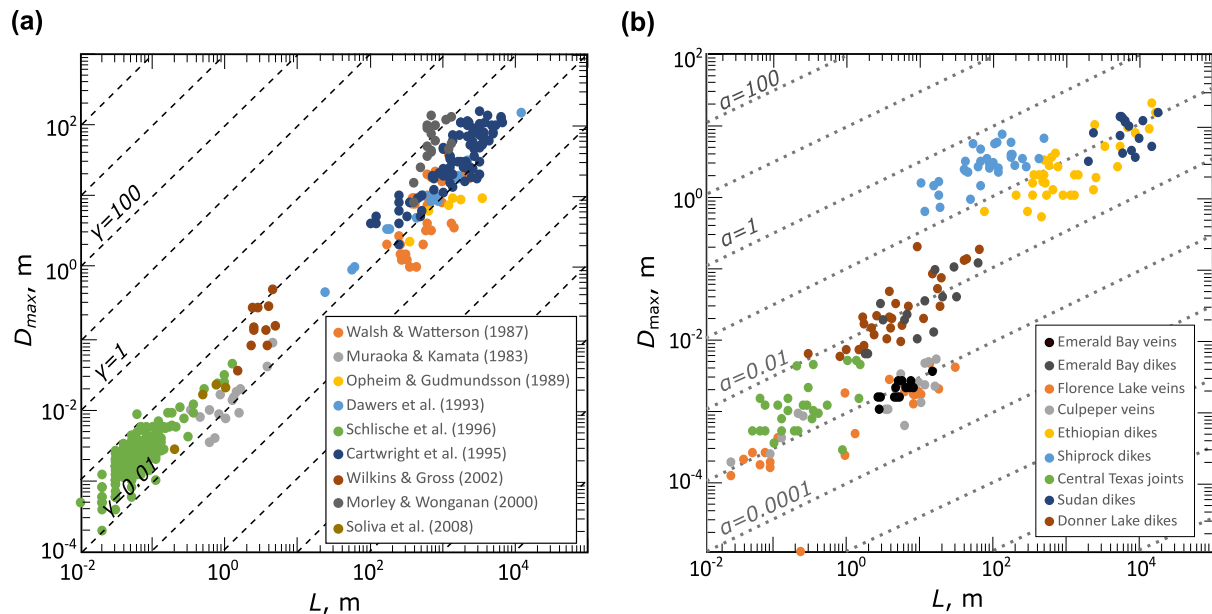


Figure 1. Compilation of observed $D_{\max}/L$ scaling of normal faults and opening-mode fractures. (a) Populations of normal faults showing maximum displacement as a function of their lengths. For reference, linear power-law functions and different γ values (Equation 1) are plotted as black dashed lines. Normal fault data sets are from Cartwright et al. (1995), Dawers et al. (1993), Morley and Wonganan (2000), Muraoka and Kamata (1983), Opheim and Gudmundsson (1989), Schlische et al. (1996), Soliva et al. (2008), Walsh and Watterson (1987), and Wilkins and Gross (2002). (b) Populations of opening-mode fractures showing maximum apertures as a function of their lengths. For reference, the sublinear power-law functions with the exponent of 0.5 and different α values (Equation 2) are plotted as gray dotted lines. Opening-mode fracture data are compiled from Vermilye and Scholz (1995): Florence Lake veins, Culpeper veins; Schultz et al. (2008): Ethiopian dikes; Delaney and Pollard (1981): Shiprock dikes; Moros (1999): Central Texas joints; Babiker and Gudmundsson (2004): Sudan dikes; and Klimczak et al. (2010): Emerald Bay veins, Emerald Bay dikes, Donner Lake dikes.

bands predominantly showing shear strain and sublinear scaling for deformation bands having a sufficiently large component of normal strain observed across them.

The $D_{\max}/L$ scaling for mixed-mode fractures involving opening-mode displacement is least understood. While disaggregation bands—deformation bands in porous sandstones showing shear strain with porosity increases across them—received some attention (Schultz et al., 2008, 2013), no fracture scaling has been established for larger structures in rift zones that commonly exhibit sheared joints and normal faults with opening displacement components. Given the differences in scaling behavior observed between normal faults (Figure 1a) and opening-mode fractures (Figure 1b), identifying and distinguishing between the displacement components of both modes of fracturing is key to resolving the scaling behavior of this type of mixed-mode fracture.

1.3. Goal and Significance of the Study

$D_{\max}/L$ scaling is applied widely in structural geology and tectonics to calculate tectonic strains (e.g., Nahm & Schultz, 2011), study fault growth, segmentation, and linkage (e.g., Cowie et al., 1993; Hughes & Shaw, 2014), infer geometric and mechanical properties of the host rock (e.g., Dawers et al., 1993; Soliva et al., 2005), and to model fluid flow through fracture networks (Klimczak et al., 2010). Determining a $D_{\max}/L$ scaling relationship for mixed-mode fractures showing opening and shear displacement would add to the understanding of all of the above aspects and have implications for tectonics at large rift systems on Earth and other planetary bodies.

To understand the evolution of mixed-mode displacements and lengths, we conducted structural fieldwork at the mixed-mode Koa'e Fault Zone. The Koa'e Fault Zone is well-preserved, and the length scales of fractures in this zone range over a few orders of magnitude, making it an ideal site. While previous work has been conducted at fault exposures of various structural segments (e.g., Bubeck et al., 2018; Denlinger & Okubo, 1995; Ge et al., 2019; Holland et al., 2006; Martel & Langley, 2006), continuous and systematic measurement of displacements along the entire fault zone and inclusion of fractures with a wide variety of lengths has not been carried out before but is essential for our understanding of $D_{\max}/L$ scaling. We collected field measurements, observed structural relationships, and compared them with previous studies to develop a conceptual fault

evolution model along the Koa'e Fault Zone. Such conceptual model is important for interpreting the meaning of displacement measurements and thus establishing a $D_{\max}/L$ scaling relationship, as this fault zone is structurally very complex.

2. Geologic Setting

The Koa'e Fault Zone at Hawai'i Volcanoes National Park, Hawai'i, is an active ~12-km long, ~3-km wide, SW-trending normal fault zone (e.g., Denlinger & Okubo, 1995; Ge et al., 2019; Peacock & Parfitt, 2002) located between the Southwestern and East Rift Zones that join at the Kīlauea edifice (Figure 2a). There are three major faults in the system: the Kulanaokuaiki Fault, the Ohale Fault, and an unnamed fault (Figure 2b), and these faults are accompanied by numerous subparallel smaller faults, including the White Rabbit Fault, and joints (Holland et al., 2006; Martel & Langley, 2006). The normal faults have scarps mostly facing toward Kīlauea with reliefs of as much as 25 m and a dilational component of up to several meters wide (Figure 3a; Holland et al., 2006). All of the major faults have a fractured slab-like ramp structure, connecting the footwall and hanging wall with a joint separating a subvertical fault scarp (Figure 3b) and a monocline (Figure 3c; also see Figure 3d). This ramp structure is often referred to as a monocline or monoclinical bridge if there is a cavity underneath the uppermost lava flow. Early studies suggested that faults are formed by en echelon joints at the surface, which propagated down and transition to faults and warp into a monoclinical flexure (e.g., Cartwright et al., 1995; Cartwright & Mansfield, 1998; Duffield, 1975; McGill & Stromquist, 1979). More recent studies argued that subsurface faults propagate upwards to form the monoclinical flexure, cavities at the fault tips, and joints at the surface (e.g., Holland et al., 2006; Kaven & Martel, 2007; Martel & Langley, 2006; Podolsky & Roberts, 2008).

The Koa'e Fault Zone is developed at the southern flank of the Kīlauea volcano (Figure 2a) with the hanging wall sliding to the north toward the edifice. The fault slip is mostly caused by the weight of the volcanic island. There are historical shallow earthquakes that occurred along the Koa'e Fault Zone (Chen et al., 2019; Ge et al., 2019). Geologic observations indicate that the current scarp and all associated structures have formed after the cooling of the surface lava flow, with the most recent resurfacing event occurring ~400–750 years ago (Duffield, 1975; Wolfe & Morris, 1996), supported by radiocarbon and palaeomagnetic methods (Holcomb, 1987). The age of the Koa'e Fault Zone has been estimated to be ~1,000 years old owing to its subdued topographic expression (e.g., Duffield, 1975), but recent studies suggest an older age as lava flows likely superposed pre-existing manifestations of the faults (e.g., Day et al., 2005).

3. Field Methods

We conducted structural field mapping along the Koa'e Fault Zone, focusing on the Kulanaokuaiki, Ohale, and White Rabbit Faults, and their associated joints and sheared joints. We mapped on the high-resolution satellite photograph of the Koa'e Fault Zone, taken by 2023 Airbus, Centre National D'Etudes Spatiales, that was printed at a scale of 1:10,000 with geologic observations recorded as structural descriptions in our field notes, field sketches, and by taking photographs. We measured fracture orientations, the relief and opening displacement components (Figure 3a), fracture lengths, and widths and heights of monoclines where they were present (Figures 3b–3d) at field localities marked with GPS recordings. Measurements were taken with approximately equal spacing of field localities along the major faults and their two associated unnamed faults with monoclines. We identified and recorded the maximum displacements and corresponding lengths of the associated joints. Measurements were taken using handheld tapes on smaller exposures and a *TruPulse 200X* laser rangefinder on larger exposures. The accuracy of the tape is ± 0.05 cm, and that of the laser rangefinder is ± 4 cm. The displacement measurements were taken from the hanging walls of the faults. Joints were measured at offset geological markers, where the shapes of fracture surfaces or lava flows matched on both sides of the fracture wall, showing minimal erosion or slumping (see white dashed lines in Figure 3a) and thus indicating how the two fracture sides connect. The fracture orientations were measured with a Brunton compass.

4. Structural Observations Across the Koa'e Fault Zone

We collected structural measurements at 233 localities along the fault zone and produced a structural map as a basis for further analysis. Our structural map (Figure 4) presents these stations along the accessible exposures of the main faults of the Koa'e Fault Zone. The Kulanaokuaiki, Ohale, and White Rabbit Faults all have fault scarps with an opening component and a monocline developed along the entire or partial length of the fault (Figure 4).

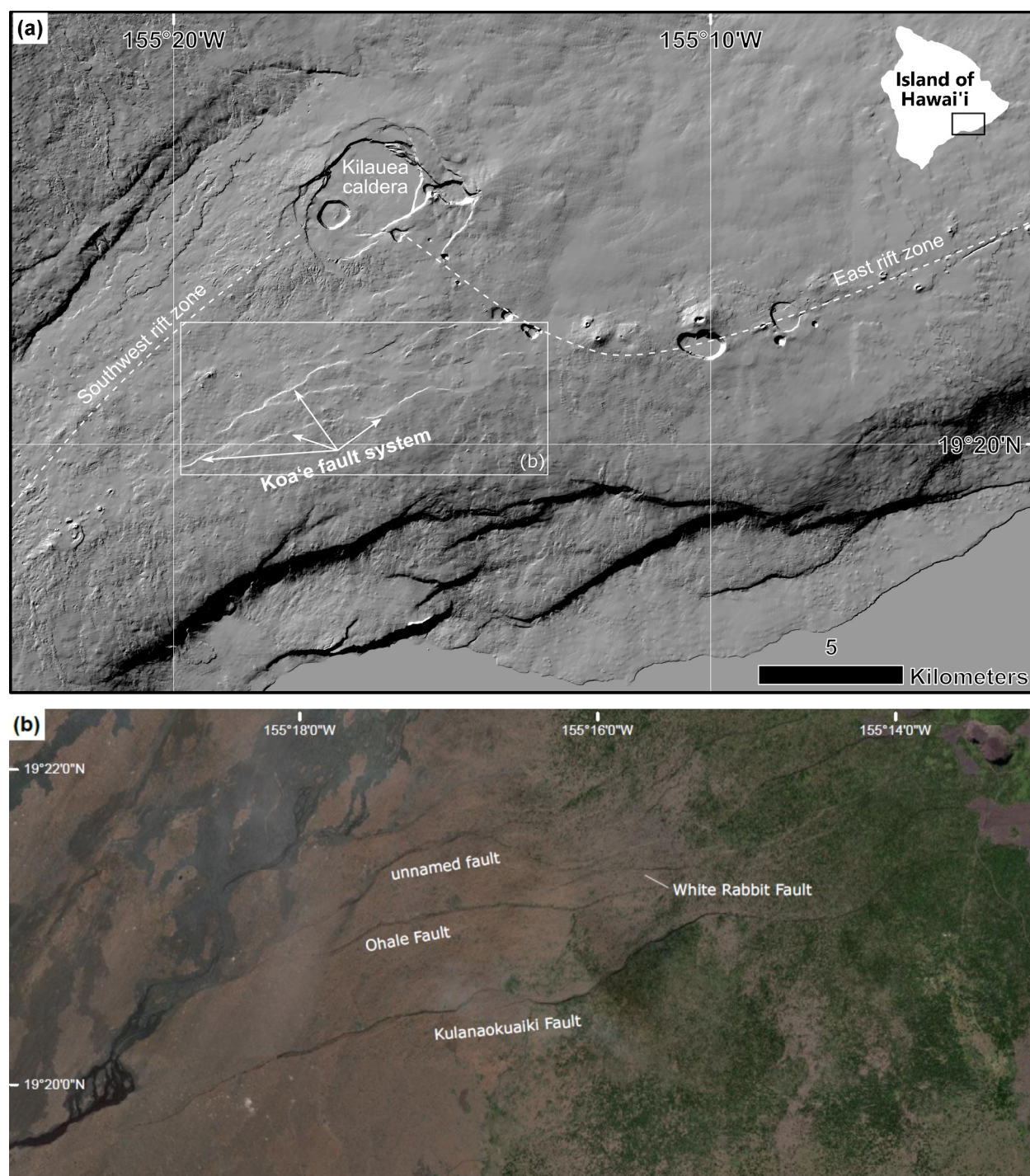


Figure 2. Overview of the Koa'e Fault Zone located in Hawai'i Volcanoes National Park, Hawaii. (a) Shaded-relief location map of the east and southwest rift zone merging at the Kilauea caldera at the Hawai'i Volcanoes National Park, Hawaii. The Koa'e Fault Zone is located at the south of the caldera in between the two rift zones. The upper right insert shows the location on the Island of Hawai'i. (b) High-resolution satellite image of the Koa'e Fault Zone, taken by 2023 Airbus, Centre National D'Etudes Spatiales. The Kulauea Fault, Ohale Fault, White Rabbit Fault, and an unnamed fault, labeled in white, show noticeable scarps in the photogeology.

Two minor faults also have monoclines, but their scarps face south, opposite to the other faults. One of the minor faults is located south of the Ohale Fault (Fwm 1 in Figure 4), and the other is located northeast of the White Rabbit Fault (Fwm 2 in Figure 4). Numerous joints are associated with and subparallel to the fault zone (Figure 4).

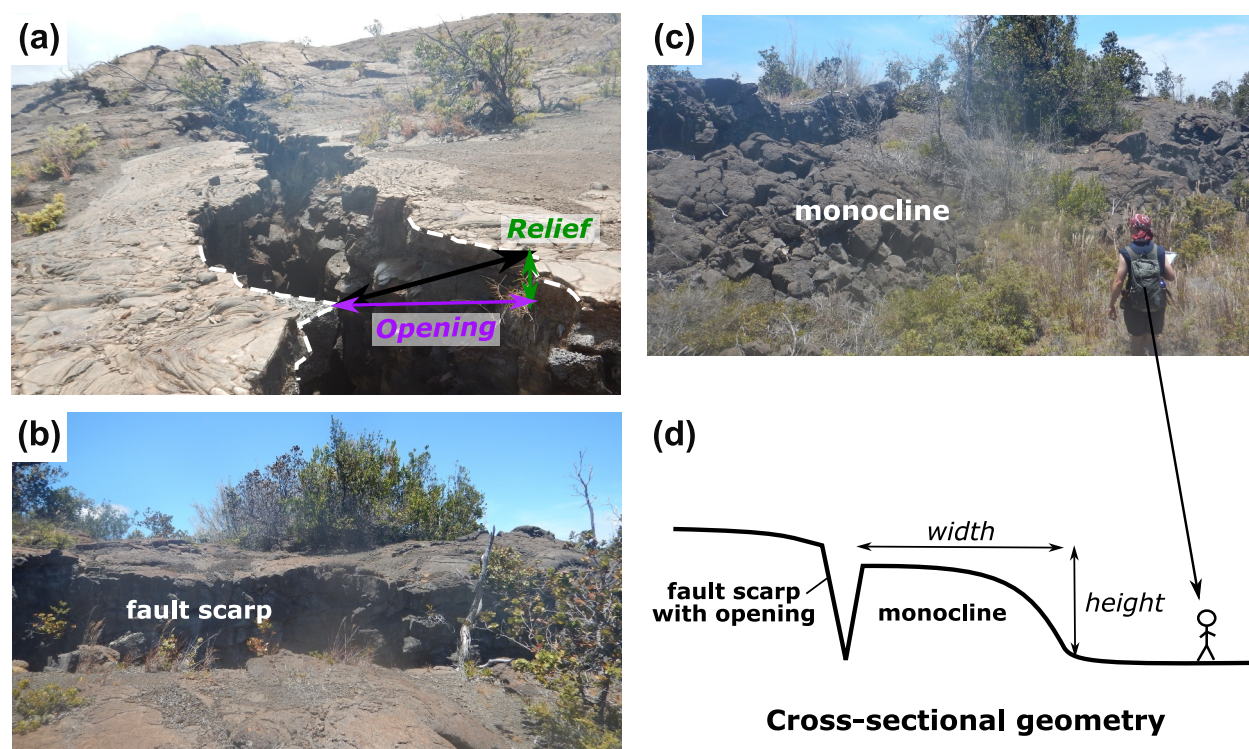


Figure 3. Field photographs of the faults and associated structures. (a) Photograph of a fracture showing vertical relief and opening. The different displacement components are shown as arrows representing relief (green arrow), opening (purple), and total displacement (black). The matching sides of the fracture wall are outlined with white dashed lines. (b) Photograph of the White Rabbit Fault showing the fault scarp and opening as seen from the top of the monocline. (c) Photograph of the White Rabbit Fault depicting the monocline and fault scarp as seen from the base of the monocline. Both views in (b) and (c) are toward the south, and the fault scarp is facing to the north. (d) A schematic cross-sectional geometry of the fault shows the spatial relationship of the fault scarp, opening displacement, and width and height of the monocline.

To distinguish these structures from the pre-existing basaltic columnar joints across the study area, we use the term ‘joints’ (i.e., ‘pure joints’ and ‘sheared joints’) only when referring to the fault-parallel structures, whereas ‘columnar joints’ specifies the pre-existing structures of various orientations that are inherent to basalts. While the unnamed major fault to the north of the Ohale Fault is accessible for field mapping, the base of the hanging wall is covered with a thick layer of sand, hindering geological observations and measurements.

4.1. Faults

4.1.1. Kulanaokuaiki Fault

The Kulanaokuaiki Fault is approximately 12 km long, and we collected data along 8.6 km of it from the western tip toward the east. Beyond this, for about 3 km, the fault is covered by dense vegetation and thus is inaccessible. Three mapped fault segments are separated by relay ramps. The western tip of the fault ends in a zone of en echelon jointing that extends for another ~1 km, which we interpret as part of the fault damage zone. Subparallel pure and sheared joints accompany the Kulanaokuaiki Fault, with some minor faults being synthetic and others being antithetic to it (Figure 4). Pits with widths that exceed 10 m are located near the center of the western segment of the Kulanaokuaiki Fault.

The Kulanaokuaiki Fault has a subvertical fault scarp that utilizes preexisting columnar joints as indicated by its jagged edge (Figure 5a). A monocline is found directly adjacent to the fault scarp (Figure 5a). It is best developed on the western fault segment (Figure 4). The monocline is highly fractured, showing an abundance of hexagonal patterns from pre-existing columnar joints. No opening or dilational components of the fault were directly observed and thus could not be measured, but the collapse of the monocline was present in many places, indicating that any voids were filled in. In places, only the interior of the monocline collapsed into the void, leaving behind a natural bridge in the shape of the monocline formed by the uppermost basaltic layer (Figure 5b). A cross-sectional

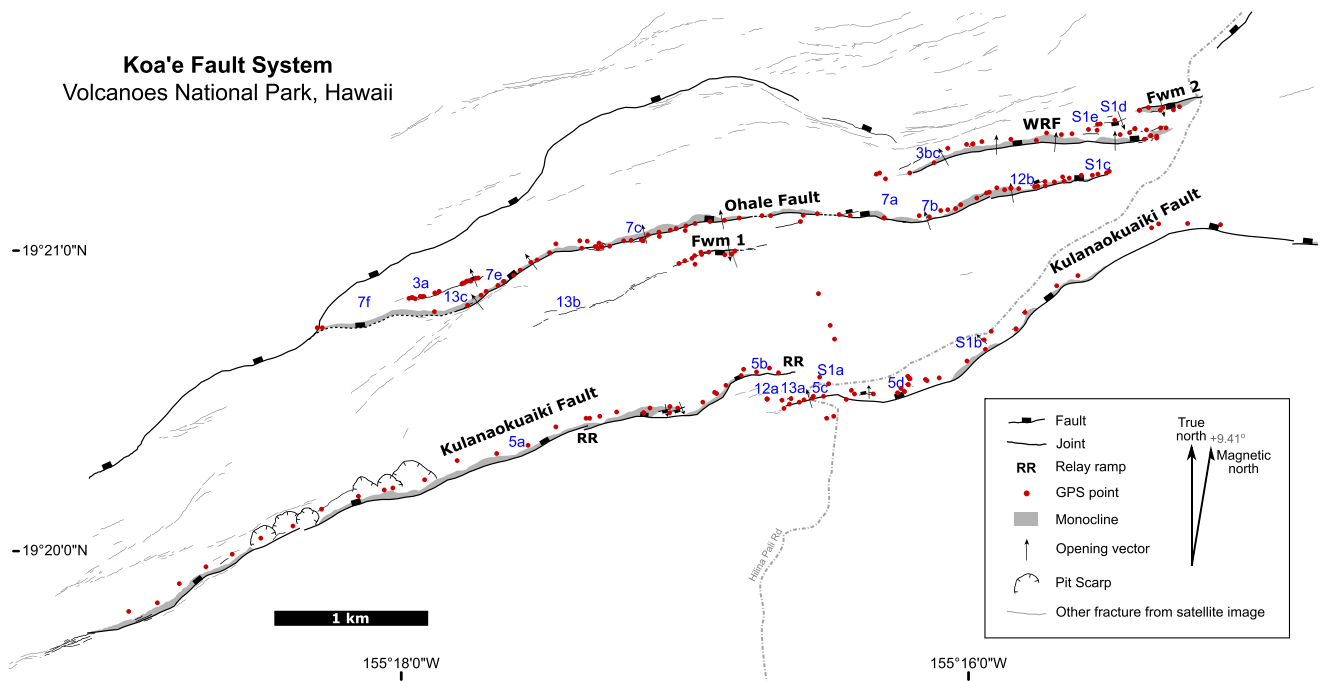


Figure 4. Structural map of the Koa'e Fault Zone. Blue labels correspond locations of photographs in Figures. The eastern part of the Kulanaokuaiki Fault continues for another ~3 km to the east. WRF = White Rabbit Fault. Fwm = Fault with monocline. The GPS points are labeled in red dots, and their structural measurements are listed in Table 1 and the supplementary materials.

view of the fault is exposed at the Hilina Pali Road cut, which coincides with one of the relay ramps of the fault (Figures 4 and 5c). Here, the monocline collapsed along a 5.8-m-wide swath directly adjacent to the fault scarp, but some remnants of the monocline show the flexure of the structure (Figure 5c). A large portion of the eastern fault segment lacks a monocline and only has the subvertical fault scarp (Figures 4 and 5d). The scarp is highly irregular due to collapse along vertical columnar joints and horizontal basaltic layers, with the talus and scree obscuring the lower part of the scarp (Figure 5d). The hanging wall also exhibits hexagonal columnar joints (Figure 5d). Distinctly larger joints that are not part of but utilize the columnar jointing are subparallel to the major fault scarp, showing a zigzag pattern of the opening displacement (Figure 5d).

With the monocline collapsing and filling the opening of the fault (Figures 5a and 5d), we were only able to measure the total reliefs, that is, the vertical distance between the top of the footwall to the hanging wall. The relief profile has a general plateaued shape (Figure 6) that is typical of fault segment interactions, as evident by the relay ramps observed in the field (see Figure 4). The maximum relief of the middle segment is located at the center of the profile, with that of the west and east segments also skewed towards the center.

4.1.2. Ohale Fault

The Ohale Fault was mapped along its 5.4 km of length from the eastern tip toward the west, where it merges with the unnamed fault (Figure 4). A monocline is present along almost the entire fault scarp (Figure 7a), except for a few locations (Figures 4 and 7b). At the western part of the Ohale Fault, numerous examples of well-preserved pahoehoe lava are observed to have a paleo-flow direction that points up-slope on the monocline (Figure 7c), indicating that the terrain was flat or sloped in the opposite direction at the time of flow emplacement and that the relief of the monocline and fault scarp formed entirely after the most recent lava flows had cooled. The fault scarp is highly fragmented by preexisting vertical columnar joints and horizontal basaltic flow contacts. Both the fault scarp and monocline are absent in two locations near the center of the Ohale Fault, but joints are observed to continue along the strike of the scarp and connect to the next fault scarp (dashed line in Figure 4), which is indicative of fault linkage between several fault segments.

Subparallel joints, a minor fault, and small pits accompany the Ohale Fault on the hanging wall (Figure 7d) and footwall (Figure 7e). The minor fault that also shows a monocline is 314 m long and is south of the Ohale Fault,

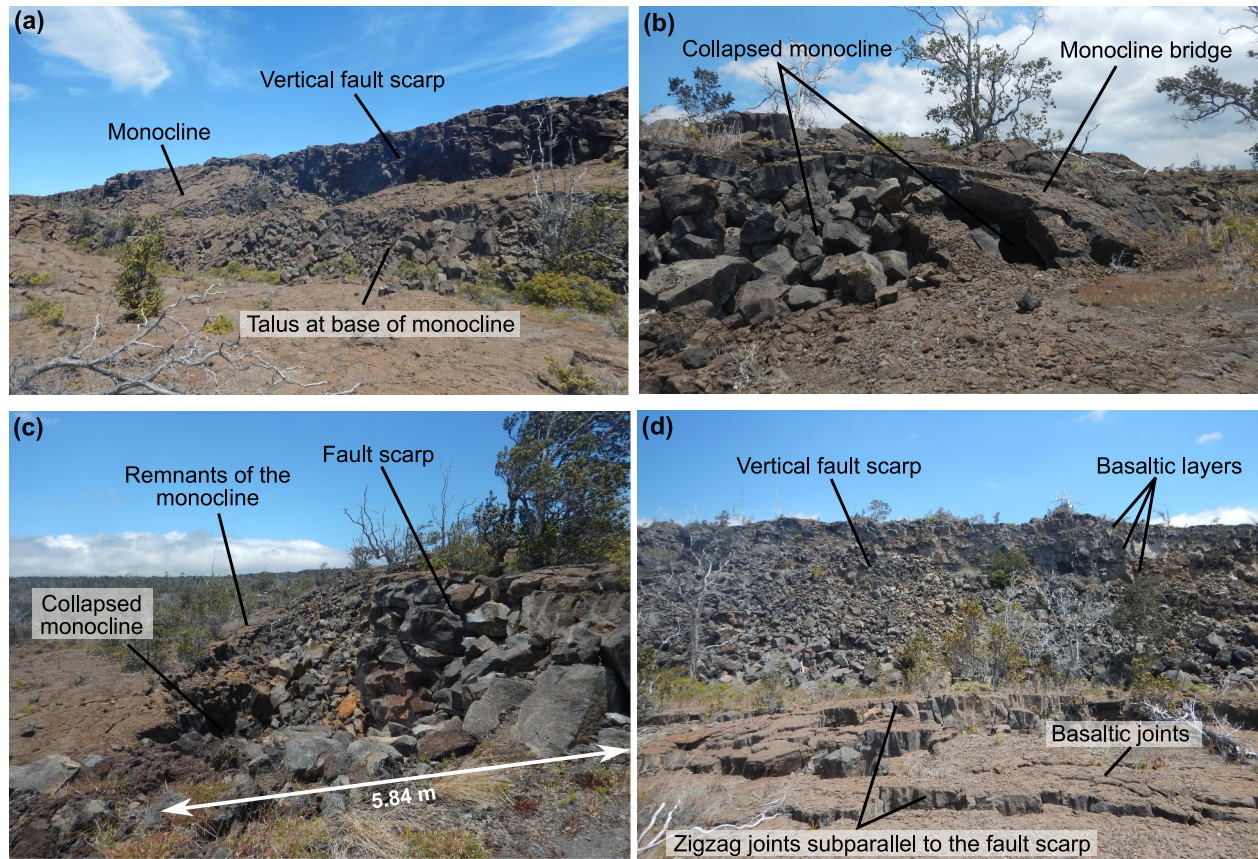


Figure 5. Field observations at the scarp of the Kulanaokuaiki Fault. (a) Subvertical fault scarps with monocline. The view is toward the southeast. (b) The interior of the monocline has collapsed, leaving a monoclinical bridge. The view is toward the southwest. (c) Cross-section view of the Kulanaokuaiki Fault at the Hilina Pali Roadcut. The view is toward the east. (d) Vertical fault scarp with subparallel joints. The view is toward the south. See locations in Figure 4.

where it connects to a set of echelon joints on both tips (Figure 4). The scarp of a large, unnamed fault merges with the western tip of the Ohale Fault where they form a prominent fault scarp that extends to the west of this intersection (Figures 4 and 7f). The base of the unnamed fault is covered with a thick layer of black sand; thus, its relief and structures on the hanging wall could not be accurately measured.

The morphology of Ohale Faults is dominated by the width and height of the monocline (Figure 8). The relief and the opening are only pronounced where there are no monoclines. No relay ramps are observed in the field. However, the plateau-shaped profiles indicate some fault segment interactions, which may be at locations with no monoclines or faults and where only joints and slumping are observed (i.e., inferred fault in Figure 4). The overall displacement of the faults and monocline is lower when there is a large joint at the footwall, which accommodates some of the displacement (Figure 8). The minor fault with monocline associated with the Ohale Fault (Fwm 1; see

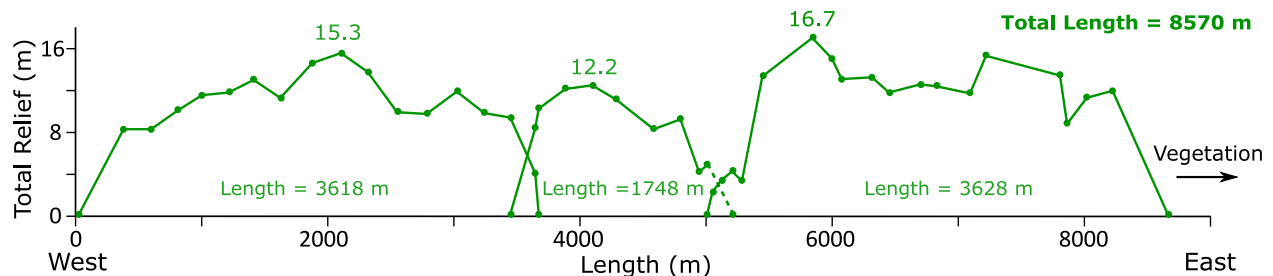


Figure 6. Relief profile of the Kulanaokuaiki Fault with 45 measurements plotted along the fault length at a spacing of ~200 m. The maximum reliefs for each fault segment are identified and labeled along the profile with lengths. Relay ramps are presented in the profiles at the locations showing overlapping reliefs.

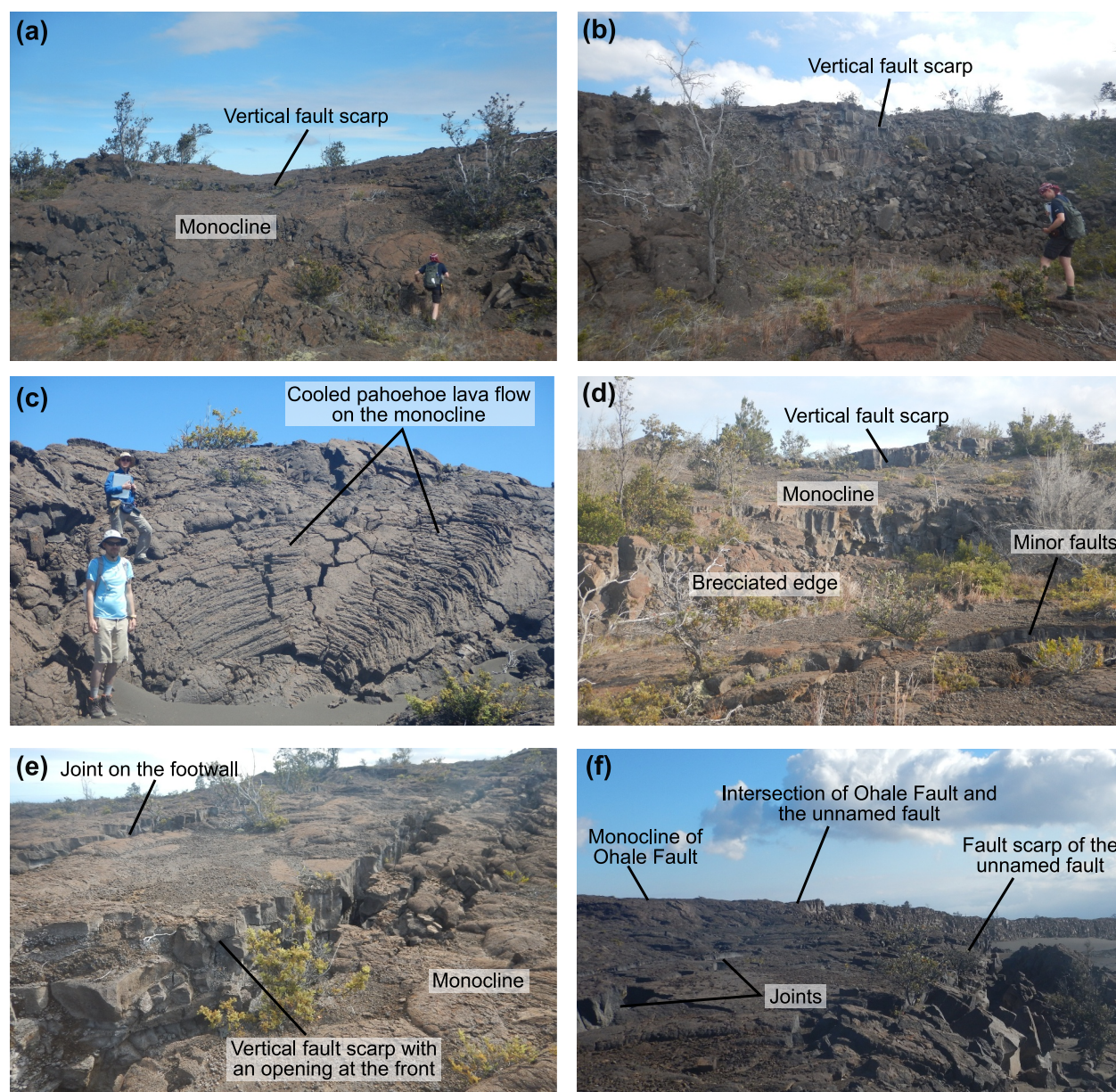


Figure 7. Photographs of structural relationships at the Ohale Fault. (a) Vertical fault scarp with monocline. The view is toward the south. (b) Vertical fault scarps with no monocline. The view is toward the south. (c) Example of a tilted Pahoehoe lava flow surface where the flow direction points upslope of the monocline. The view is toward the southeast. (d) View toward the south showing the minor fault located in the hanging wall. (e) Opening in front of the fault scarp with a large subparallel joint on the footwall, view from the top of the monocline toward the southeast. (f) The intersection of the western tip of the Ohale Fault and the unnamed fault, with the view direction to the west. Figure locations are shown in Figure 4.

Figure 4) exhibits a peak-shaped profile, dominated by the width of the monocline (Figure 9), rather than the plateaued shape of the larger structures (Figures 6 and 8).

4.1.3. White Rabbit Fault

The White Rabbit Fault is 2.1 km long, and a monocline is present nearly along the entire length of the fault (Figures 3c and 4) and extends beyond the fault scarp at one tip (Figure 4). The fault has similar characteristics to the Ohale Fault, including the highly fragmented fault scarp, joints, and minor faults parallel to the major structure (Figure 4). A minor fault with a monocline at the northeast of the White Rabbit Fault has a length of 420 m (Figure 4), and its scarp is facing south to southeast, antithetical to the White Rabbit Fault.

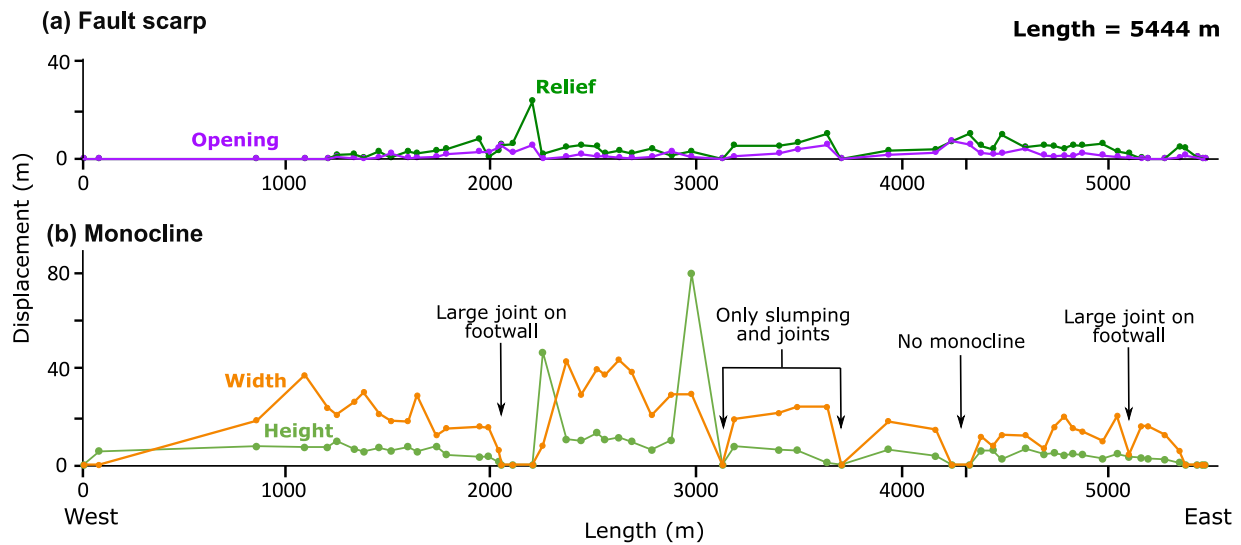


Figure 8. Displacement profile of the Ohale Fault with measurements from 60 localities. (a) Displacements of the fault scarp shown for openings in purple and reliefs in green. (b) Measurements of the monocline height are depicted in light green, and the variations in width are highlighted in orange. Locations of particular interest, such as the lack of the monocline and where large joints on the footwall occur, are labeled. The diagrams in (a) and (b) are on the same vertical scale for comparison.

Similar to the Ohale Fault, the White Rabbit Fault also has the width and height of its monocline be more dominant compared to the opening and relief of the fault scarp (Figure 10). The differences are less pronounced compared to the Ohale Fault, especially where large openings of up to ~20 m are found along the structure. No relay ramps are observed in the field, but the multiple peaks of opening and relief (Figure 10a) indicate that the fault is segmented and has experienced multiple episodes of slip along different segments. The minor fault with monocline associated with the White Rabbit Fault (Fwm 2; see Figure 4) has a morphology dominated by opening and relief of the fault scarp, which differs from the larger faults (Figures 8 and 10). The profiles of the fault scarp exhibit a relatively peak-shaped profile with plateau-shaped monocline profiles (Figure 11).

4.2. Joints

Joints are widespread across the Koa'e Fault Zone. We observe pure and sheared joints. Sheared joints—structures showing both opening and vertical relief with the opening being larger than the relief—are found to accompany and strike subparallel to the main faults (e.g., Figures 4 and 12). Preexisting columnar joints heavily influence their map patterns, resulting in zigzag patterns (Figure 12). The examples of larger sheared joints in Figure 3a and smaller ones in Figure 12 are representative of all observed sheared joints at the field site.

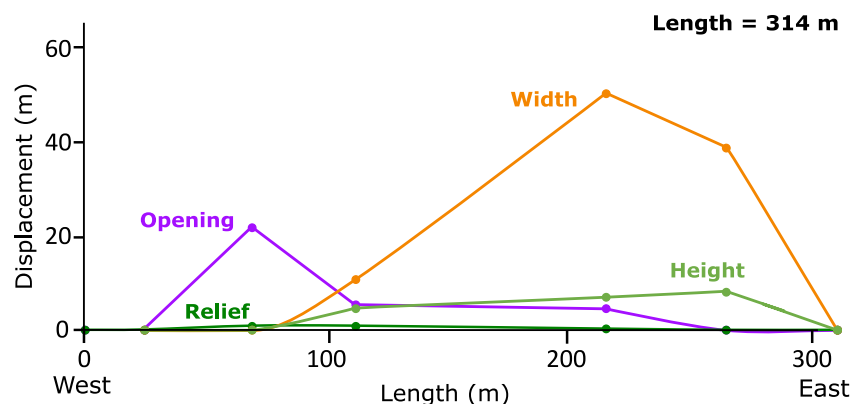


Figure 9. Displacement profile of the fault with monocline 1 (Fwm 1) that is associated with the Ohale Fault. The measurements were taken from 8 localities. The fault scarp relief is plotted in green, and the opening is plotted in purple. The monocline height is plotted in light green, and the width is plotted in orange.

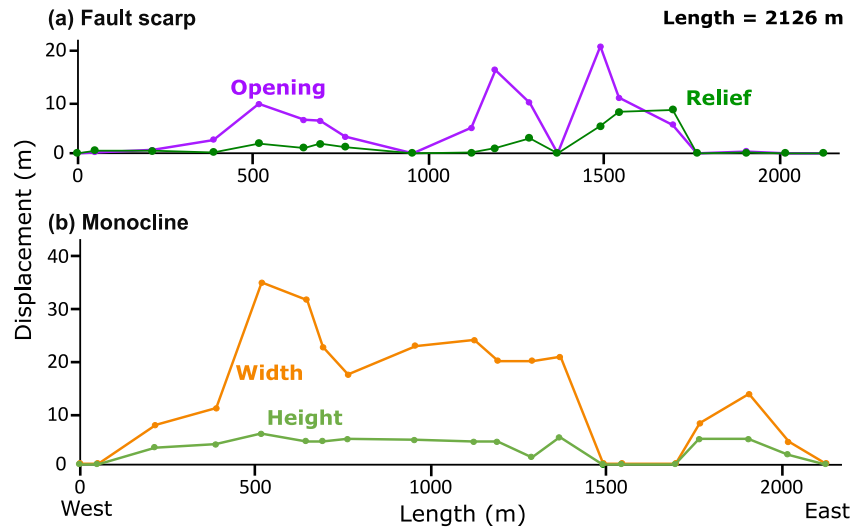


Figure 10. Displacement distribution along the length of the White Rabbit Fault with measurements from 20 localities. (a) Displacements of the fault scarp showing opening in purple and relief in green. (b) Measurements of the monocline showing the height in light green and width in orange. The diagrams in (a) and (b) are displayed at the same vertical scale for comparison.

Pure joints—structures only showing opening displacements—strike northeast to southwest, parallel to the main faults (Figure 4; see example in Figures 5d and 7e). Pure joints occur from small structures of only a few tens of centimeters to up to 200 m in length. Some of them are isolated as individual structures while others are in close association with the faults. The latter of those pure joints frequently occur in en echelon patterns at or near the tips of faults or sheared joints, such as the small-scale example in Figure 13a and the large-scale western tip of the Kulanaokuaiki Fault (Figure 4). Large joints have a fragmented and jagged appearance, similar to that of fault scarps, showing highly fractured subvertical walls influenced by preexisting columnar joints (Figure 13b). Preexisting columnar joints also heavily influence the map patterns of joints, resulting in zigzag patterns, especially for larger joints (Figures 13b and 13c; see also Figures 5d, 7e, and 7f). En echelon patterns and overlapping joints are common for small (Figure 13a) and large structures (Figure 13c) with total lengths of the en echelon joint arrays in places exceeding 1 km. All large joints crosscut lava flows (Figure 13b), and no lava is observed to have flowed into the opening. This is consistent with the observations at the faults (Figure 7c), where deformation entirely post-dates the emplacement and cooling of the most recent lava flows.

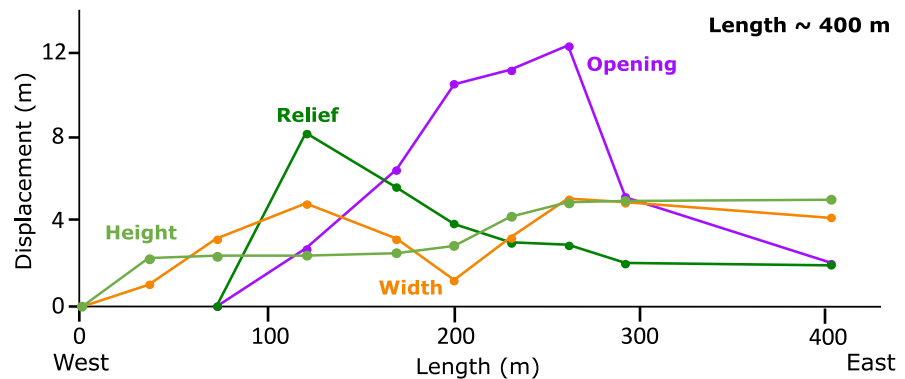


Figure 11. Displacement profile of the fault with monocline 2 (Fwm 2), associated with the White Rabbit Fault. The measurements were taken from 10 localities. The fault scarp relief is plotted in green, and the opening is plotted in purple. The monocline height is plotted in light green, and the width is plotted in orange. The eastern fault tip is clearly defined in the field beyond 400 m of length of the structure.

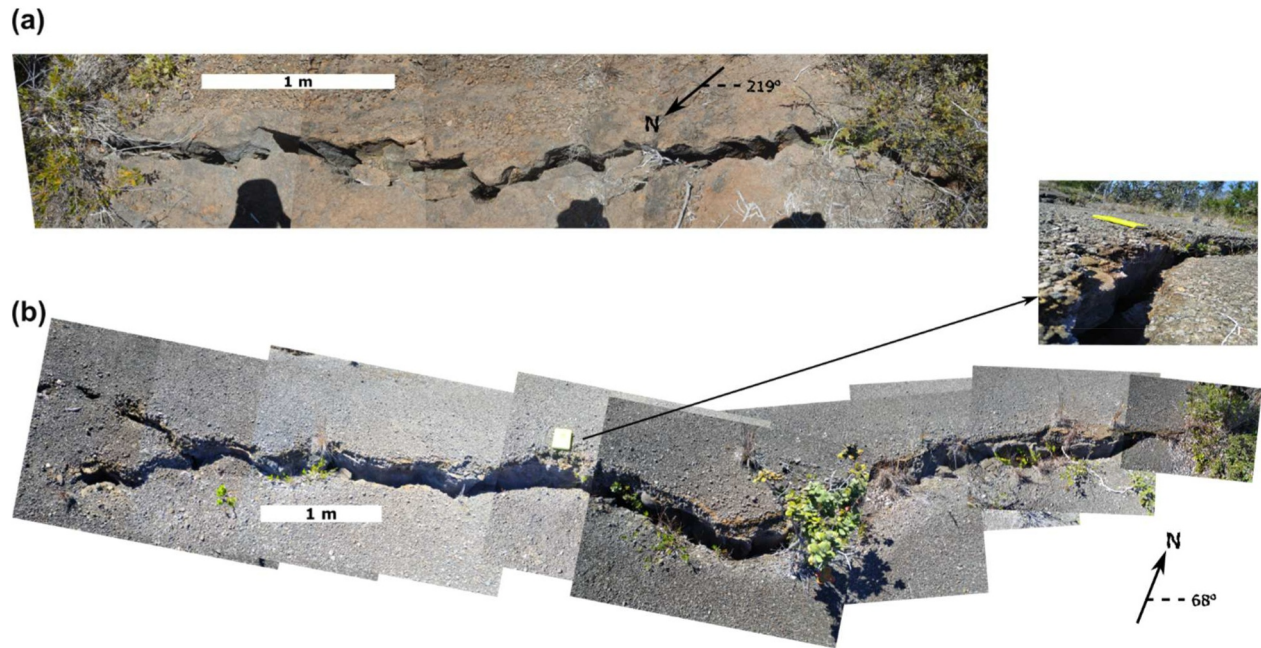


Figure 12. Field pictures of two sheared joint examples from the top view. (a) Sheared joint with a length of 4.61 m. (b) Sheared joint with a length of 10.56 m and a side-view at the right. The locations of the structures are labeled in Figure 4.

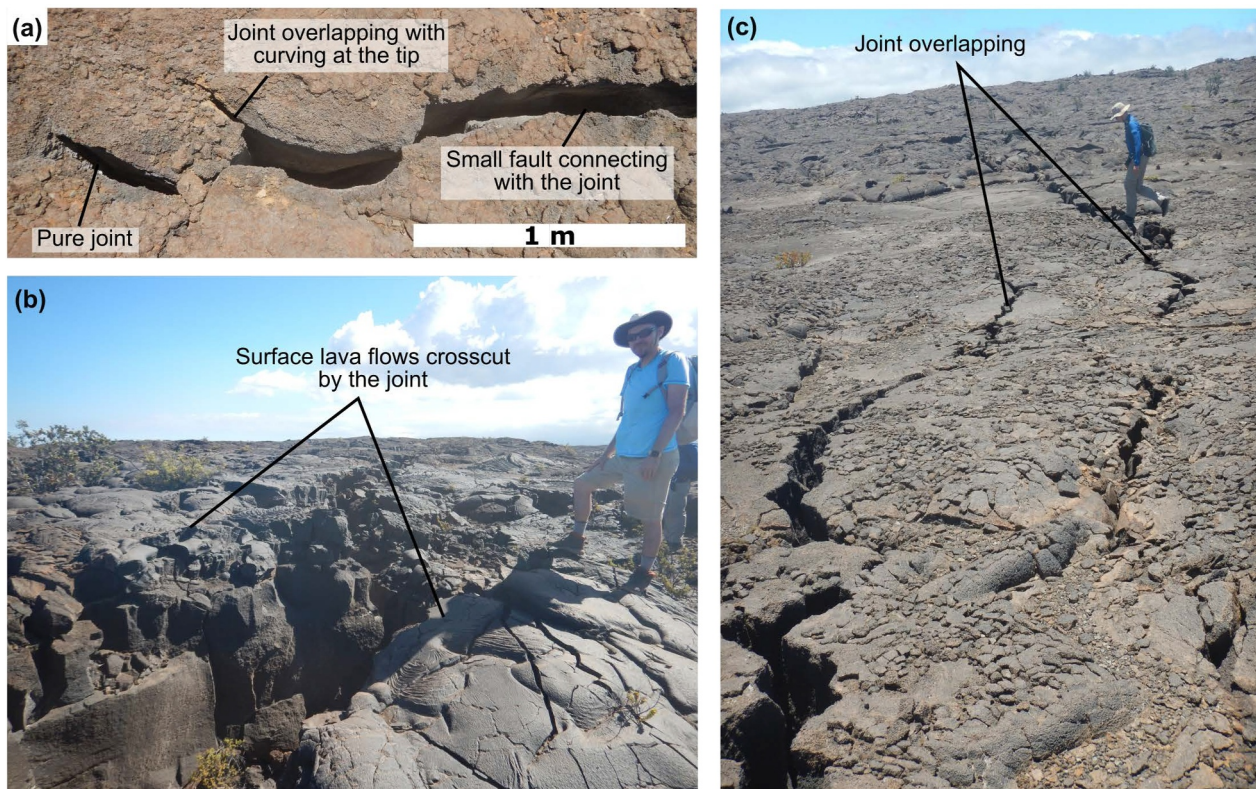


Figure 13. Examples of joints observed in the field. (a) En echelon joints observed at the tip of a fault. (b) Large joints separate lava flow and expose the basaltic columnar joints on the walls. (c) Two large overlapping joints as part of an en echelon joint array south of the Ohale Fault (Figure 4). The locations are labeled in Figure 4.

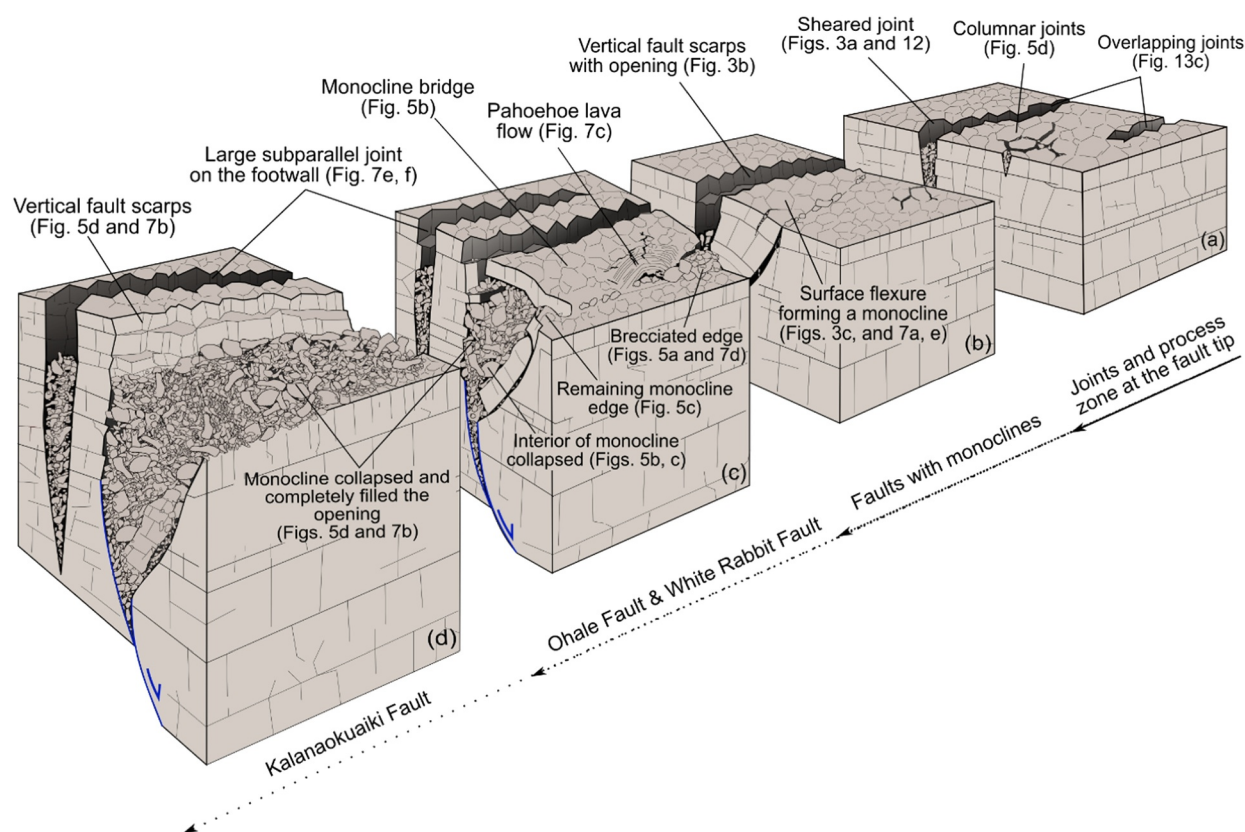


Figure 14. Block diagrams representing a morphological continuum of fault zone development of the Koa'e Fault Zone based on the most common field observations along the structures (Figure 4). All structures formed after the emplacement of the most recent lava flow. (a) The smallest structures or tips of larger faults consist of joints that form at the surface utilizing pre-existing basaltic columnar joints. Some of the joints are sheared, showing an opening and minute structural relief. (b) Near the fault tips of large faults or at minor faults there is gentle flexure in the hanging wall, forming a small monocline. The fault opening between a subvertical scarp and the monocline widens toward the center of the fault. (c) Closer towards the center of larger faults, the opening further increases, and the interior of the monocline becomes unstable and partially collapses into the opening of the fault, in places leaving monoclinical bridges with brecciated edges. Large subparallel joints are commonly found on the footwall. (d) At the center of the largest faults, the monocline fully collapses, leaving only remnants of the monocline entirely filling the opening at the base of the fault scarp. The fault slip at depth is highlighted in blue. The columnar joints are interpreted to extend at depth with the fault structure, but their actual extent and variations remain unknown.

5. Morphological Continuum of the Koa'e Fault Zone

We combine our field observations into a series of block diagrams (Figure 14), representing a continuum of geomorphological characteristics that developed along the faults as they evolved from joints and minor faults into faults with sizeable scarps. The entire morphological continuum is observed from tip to center along the biggest structures, but also individual geomorphological characteristics are observed at structures of different sizes across the Koa'e Fault Zone. The smallest structures may only display characteristics shown in Figure 13 that include subvertical joints that form at the surface utilizing pre-existing basaltic columnar joints with some joints being sheared, creating an opening and minute structural relief (Figure 14a). Some of these joints stay isolated, such as the structures shown in Figure 12, while others are commonly seen at the tips of main faults forming the process or damage zone.

Near the tips of the larger faults or at the observed minor faults there is gentle flexure in the hanging wall, forming a small monocline (Figure 14b). The topographic expression of the monocline in this stage is subtle, and it remains largely intact as a tilted block. It is well observed at the Ohale Fault (Figure 7a), White Rabbit Fault (Figure 3c), and their two associated minor faults with monoclines. The opening component of the faults that are commonly found between the vertical fault scarp and the monocline exhibits zigzag patterns, influenced by preexisting columnar joints, as is observed for all the joints (Figures 12 and 13).

Closer to the center of larger faults with monoclines, the opening widens under the fault dilation. Subsurface materials are crushed along the faults and filled in the dilation (Figure 14c), removing the support for overlying basaltic layers. Pre-existing columnar fractures then facilitate the collapse of basaltic layers above the fault. The internal of the monocline becomes unstable, collapsing into the opening created by the dilation of the fault (Figure 14c). Frequently, slabs of the most recent, uppermost basaltic flows remain to form monoclinical bridges with brecciated edges. Large subparallel joints are observed on the footwall of the fault as isolated structures or a set of echelon joints with a zigzag pattern (Figure 7e). This configuration represents the overall morphology observed at the Ohale and White Rabbit Faults, as well as those areas of the Kulanaokuaiki Fault where a monocline is preserved.

In the center of the largest faults, the monocline fully collapses, in places leaving only remnants of the monocline behind (Figure 14d). In most places, however, no evidence of a monocline remains, and its scree and talus entirely fill the opening at the fault. This leaves a pronounced fault scarp and a talus pile at its base. This stage of the structural evolution of the Koa'e fault zone is represented by the overall morphology of the Kulanaokuaiki Fault and is beginning to manifest itself at a few locations along the Ohale Fault, as seen in Figure 7b. This entire structural evolution occurred after the emplacement of the most recent lava flows, as evidenced by the cross-cutting relationships and pahoe-hoe lava flow directions pointing up-slope on the monoclines (Figures 7c and 14c).

6. Structural Evolution Through Downward Propagation

The morphological continuum (Figure 14) represents the structural evolution of the Koa'e Fault Zone. Subvertical jointing is initiated at the surface (Figure 14a) and propagates downwards, transition to faulting at a depth where the appropriate loading conditions are reached. At that stage, joints are sheared, and the more a structure propagates downward it evolves into a minor fault. With increasing accumulation of fault slip, the displacement is accommodated over a broader region of the hanging wall, and a monoclinical flexure develops (Figure 14b). This could be due to the numerous pre-existing columnar joints accommodating the displacement cumulatively across the rock mass, resulting in more distributed deformation. As the fault plane continues to grow with non-vertical dips, a dilational displacement component that results from the sliding of the fault continues to widen the surface openings formed previously by the jointing (Figure 14c). Very similar observations of joints and normal faults are also reported from Almannagja, Iceland, where surface subvertical joints propagate downwards and become inclined faults at depth (Gudmundsson, 2017, pp. 51–58).

The interpretation of downward propagation is further supported by the fact that pure joints, sheared joints, and minor faults are found not only in association with the main faults but also in isolated locations away from the main faults. The interpretation aligns with earlier studies that support downward fault propagation (e.g., Cartwright et al., 1995; Cartwright & Mansfield, 1998; Duffield, 1975; McGill & Stromquist, 1975, 1979; Strömquist, 1976) but contradicts upward propagation models of the faults (e.g., Holland et al., 2006; Kaven & Martel, 2007; Martel & Langley, 2006; Podolsky & Roberts, 2008).

The upward propagation models, as illustrated in conceptual models by Martel and Langley (2006) and Kaven and Martel (2007), suggest that the slip of a blind fault at depth causes surface flexure in the form of a monocline, which triggers the formation of en echelon joints on the footwall following the strike of the fault. As the fault tip propagates towards the surface, the limb of the monocline steepens, and a cavity is suggested to form on the hanging wall beneath the monocline. This cavity is interpreted to either create a monoclinical bridge or collapse entirely, leaving only the fault scarp. Eventually, the upward propagation model requires the en echelon joints on the footwall to connect with the faults, forming the subvertical scarp. In the model, antithetic faulting occurs in the subsurface of the hanging wall, which then leads to the formation of joints at the surface. Accordingly, large surface joints are predicted to be formed on the footwall of the upward propagating fault and associated with the monocline, regardless of whether the fault has breached the surface or not. The upward propagation model invokes that monoclines should be present at the fault tips while the fault scarps form toward the fault center. Fault scarps without an additional opening displacement component should be observed when the fault reaches the surface and prior to connecting with joints.

We identify several geological observations that are inconsistent with the predictions of the upward propagation model described above. Pure joints, sheared joints, and minor faults are found not only in association with the main faults but also in isolated locations away from the main faults, showing that joints need not be associated with faults and monoclines. In fact, isolated joints and en echelon joints are observed subparallel to the fault zone

throughout the field site but are not associated with any monocline or surface flexures on either side of the fracture (Figures 12 and 13c). These observations are in accord with numerous joints found to be associated with major faults on the Reykjanes Peninsula in southwest Iceland, formed by absolute tensile stress at the surface (e.g., Gudmundsson, 1987a, 1987b).

Furthermore, we observe sheared joints without any associated monocline (Figure 12). Similarly, en echelon joints are commonly observed to form the process zones at the fault tips whereas a monocline is more commonly found to have formed towards the fault center, such as seen at the western tip of the Kulanaokuaiki Fault and the minor faults (Figure 4). We only observe one location at the White Rabbit Fault, where the monocline is found beyond the fault scarp, and where it interacts with the minor fault with the monocline (Fwm 2, Figure 4). This phenomenon may also be explained by the fault plane being well-developed beneath the most recent lava flows, dated to be 500–750 years old (Holcomb, 1987), where the most recent fault slip did not cause surface ruptures at this location. All these surface observations are consistent with fractures initiated at the surface as joints.

The spacing of the fractures gives a rough estimate of the depth in evolving dilational fracture systems (e.g., Forslund & Gudmundsson, 1991; Gudmundsson, 1987b; Nur, 1982). The lateral spacings of fractures along the Koa'e Fault Zone are mostly one to several hundred meters, and one to two km for the major faults (Figure 2). These spacing widths are in the same order as the depth of seismic activity (~ 2 km) recorded along the Ohale and Kulanaokuaiki Faults (Ge et al., 2019). Given that these two major faults are among the longest structures, they are expected to be the deepest in the study area, as well (Figure 2). This may indicate that pure and sheared joints show a depth range of several hundred meters and normal faults exceeding one-km depth. This transition depth is consistent with calculations of maximum joint depth using the Griffith criterion (Gudmundsson, 2011, pp. 212, 235–236). Beyond this depth, the overburden pressure shifts all tectonic stresses into a purely compressive regime, allowing for joints to be reactivated as normal faults as they propagate deeper.

7. Displacement-To-Length Scaling of the Mixed-Mode Koa'e Fault Zone

We measured the geometric characteristics, including displacements and lengths of 75 structures at 233 localities (see Figure 4). This data set is comprised of the three named faults, two unnamed faults with monoclines, 18 sheared joints, and 52 pure joints. The full set of measurements is presented in the supplementary material (Table S1 in Supporting Information S2). To perform a displacement-to-length scaling analysis, we compiled the maximum displacements of all of these structures (Table 1). We determined the maximum displacements of large structures from the displacement distribution profiles for the Kulanaokuaiki Fault (Figure 6), Ohale Fault (Figure 8), White Rabbit Fault (Figure 10), and the two unnamed faults with monoclines (Figures 9 and 11). We also obtained the displacement distribution profiles for five sheared joints (Table 1, marked with asterisks; Supplementary Figure S1 in Supporting Information S1).

The mixed-mode Koa'e Fault Zone has a complex tectonic morphology showing multiple components of displacement, and therefore the different modes and their components must be clearly defined to establish a maximum displacement-to-length ($D_{\max}/L$) scaling relationship. For pure joints, their total displacements (D) are entirely represented by their apertures, A (Figure 15a). When joints propagate sufficiently deep to reach the jointing-faulting transition, shear from fault slip (S) is added to the aperture (Figure 15b). For dipping fault planes with dip angles θ , the total displacement of the mixed-mode fracture is inclined, whereby the vertical component of the fault displacement, that is, the throw, creates a relief (S_v). The horizontal component of the fault displacement, that is, the heave, widens the original joint aperture, thus adding a dilational component (S_h). The total opening that one can observe in the field is thus the sum of the initial joint aperture and the dilation from faulting ($A + S_h$). When shear displacement is just initiated, the aperture of joints dominates the total displacement of the fracture with only small contributions from dilation and relief by faulting, leading to sheared joints. As the fracture continues to propagate deeper, shear displacement from faulting increasingly contributes to the total displacement (Figure 15c). The fault dilational component continues to widen and becomes the dominant component of the opening displacement, while also substantial structural relief is built that ultimately becomes the largest displacement component. For faults with monoclines, the monocline also contributes to the total displacement, whereby the height of the monocline is caused by also the fault slip (Figure 15d). Thus, the total displacement of faults with monocline is determined by the opening ($A + S_h$) and the total relief (S_v) from the top of the fault scarp to the base of the monocline, that is, the sum of the height of the fault scarp and the monocline.

Table 1

Compilation of Lengths and Maximum Displacements of All Measured Structures, Including Faults With Monoclines, Sheared Joints, and Pure Joints

Structure	Latitude (°)	Longitude (°)	GPS #	Length	Pure joints	Sheared joints/ faults		Monoclines	
					Aperture	Opening	Relief	Height	Width
Kulanaokuaiki Fault ^a	<i>See the location in Figure 4; refer to the supplementary materials for GPS # and measurements</i>			857,043			1,677		
Ohale Fault ^a				544,364		2,425	760	8,070	4,429
White Rabbit Fault ^a				212,557		1,990	808	573	3,381
Fault w/monocline 1 ^a				31,393		2,200	82	830	5,063
Fault w/monocline 2 ^a				39,154		1,239	821	507	510
Sheared joint ^a	19.3484001	−155.295509	3	461		26	9.7		
Sheared joint	19.3483009	−155.295794	5	153		12	5.5		
Sheared joint	19.3481998	−155.296262	6	2,133		129	10.6		
Sheared joint	19.3425999	−155.270333	28	343		144	13		
Sheared joint	19.3432999	−155.279158	38	2,577		533	16		
Sheared joint ^a	19.3463993	−155.301057	44	1,412		42	10.5		
Sheared joint	19.3493004	−155.292308	54	5,303		86	734		
Sheared joint ^a	19.3491993	−155.282755	71	772		86			
Sheared joint ^a	19.3565998	−155.256462	119	1,056		22	20		
Sheared joint ^a	19.3565998	−155.256496	120	1,768		126	90		
Sheared joint	19.3568993	−155.259143	124	672		10	7		
Sheared joint	19.3537998	−155.26125	137	257		3	1		
Sheared joint	19.3535004	−155.262353	138	508		6	3		
Sheared joint	19.3563004	−155.257977	173	6,096		17	71		
Sheared joint	19.3418007	−155.278546	205	728		8	4		
Sheared joint	19.3409004	−155.288284	207	402		6	4		
Sheared joint	19.3306007	−155.314045	211	988		20	7		
Sheared joint	19.3316994	−155.312768	212	456		4	3		
Pure joint	19.3484993	−155.295358	2	212	29.3				
Pure joint	19.3483009	−155.295794	5	68	10.5				
Pure joint	19.3483009	−155.295794	5	45	6.5				
Pure joint	19.3430996	−155.280048	39	478	9.6				
Pure joint	19.3483009	−155.29589	45	303	7.4				
Pure joint	19.3479996	−155.294217	58	1,113	58				
Pure joint	19.3477001	−155.294901	59	1,049	30				
Pure joint	19.3474998	−155.295198	60	3,018	108				
Pure joint	19.3511009	−155.283314	75	347	5				
Pure joint	19.3505001	−155.28588	79	5,906	258				
Pure joint	19.3505001	−155.286891	88	2,198	31				
Pure joint	19.3514996	−155.282741	95	3,066	308				
Pure joint	19.3516998	−155.271718	107	2,561	128				
Pure joint	19.3516006	−155.270409	109	4,117	172				
Pure joint	19.3579006	−155.25498	116	9,257	95				
Pure joint	19.3579006	−155.254533	117	170.18	11				
Pure joint	19.3567009	−155.255631	118	157.48	6				
Pure joint	19.3570995	−155.258312	125	36	3.5				

Table 1
Continued

Structure	Latitude (°)	Longitude (°)	GPS #	Length	Pure joints	Sheared joints/ faults		Monoclines	
					Aperture	Opening	Relief	Height	Width
Pure joint	19.3540993	−155.260253	134	4,516	132				
Pure joint	19.3540993	−155.260253	134	3,848	101				
Pure joint	19.3540993	−155.260253	134	9,330	200				
Pure joint	19.3540993	−155.260253	134	4,516	132				
Pure joint	19.3539009	−155.260946	135	4,777	114				
Pure joint	19.3537006	−155.261817	139	1,704	20				
Pure joint	19.3535995	−155.262824	141	9,549	214				
Pure joint	19.3533993	−155.263904	143	10,752	309				
Pure joint	19.3558006	−155.266927	163	16,809	616				
Pure joint	19.3563995	−155.257445	174	17,318	135				
Pure joint	19.3563004	−155.256213	175	10,442	130				
Pure joint	19.3430004	−155.268524	185	3,538	133				
Pure joint	19.3456001	−155.264066	189	6,226	263				
Pure joint	19.3512001	−155.256171	193	882	34				
Pure joint	19.3514004	−155.255758	194	888	10				
Pure joint	19.3430004	−155.275486	202	39	1				
Pure joint	19.3425999	−155.27497	203	336	3				
Pure joint	19.3418007	−155.276412	204	151	2				
Pure joint	19.3409996	−155.285743	206	308	4				
Pure joint	19.3407001	−155.288877	208	232	2				
Pure joint	19.3367996	−155.300842	209	450	5				
Pure joint	19.3367996	−155.300842	209	25	1				
Pure joint	19.3367996	−155.300842	209	231	2				
Pure joint	19.3300991	−155.315728	210	205	3				
Pure joint	19.3300991	−155.315728	210	625	6				
Pure joint	19.3332996	−155.309697	214	92	1				
Pure joint	19.3341999	−155.308026	215	3,708	33				
Pure joint	19.3348007	−155.306167	216	7,954	71				
Pure joint	19.3365002	−155.30234	218	8,203	101				
Pure joint	19.3369007	−155.300347	219	5,363	109				
Pure joint	19.3383999	−155.296609	221	19,969	130				
Pure joint	19.3388004	−155.294285	222	11,741	101				
Pure joint	19.3402996	−155.290846	224	5,608	63				
Pure joint	19.3411007	−155.287299	226	5,541	42				
Pure joint	19.3412991	−155.283751	229	2,478	85				

Note. The Kulanaokuaiki Fault, Ohale Fault, White Rabbit Fault, two unnamed faults with monocline, and five sheared joints have detailed displacement profile measurements available in the supplementary materials. The full data set with all GPS recordings is available in the supplementary materials (Table S1 in Supporting Information S2) and as a shapefile. All measurements are listed in centimeters. ^aDetailed displacement distribution measurements collected for this structure (supplementary materials).

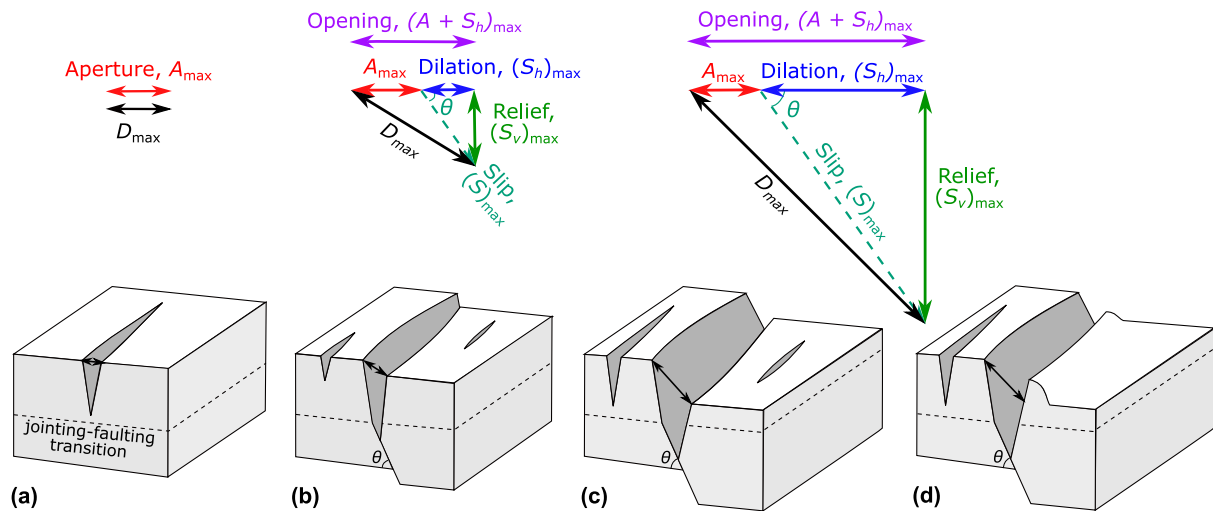


Figure 15. Block diagrams showing the growth of mixed-mode fractures with the total displacement and their vector components illustrated above. (a) Pure joint forming at the surface, in which the total displacement (black arrow) is equal to the joint aperture (A ; red arrow). (b) A sheared joint forms when the structure reaches a critical depth and faulting is initiated at an angle (θ). The total displacement is a combination of the aperture of the joint (red arrow) and the fault slip (S ; dashed green arrow). The horizontal component of the slip widens the original aperture, thus adding a dilation (S_h). The opening of the fracture is equal to the sum of the aperture from jointing and dilation from faulting ($A + S_h$). The vertical displacement component is purely formed by faulting, creating a relief (S_v). (c) When the structure propagates deeper, the mode II/III displacement dominates the total displacement of the fracture, building structural relief that exceeds the opening (i.e., $S_v > A + S_h$). (d) Monoclines contribute to the total displacement, particularly the height of the monocline affecting the vertical relief. Thus, the total displacement of faults with monocline is governed by the opening ($A + S_h$) and the total relief (S_v) from the top of the fault scarp to the base of the monocline.

We scale the maximum total displacement ($D_{\max}$) of fractures in the Koa'e Fault Zone with their lengths (L) following the methods of Clark and Cox (1996), compiling all data including those for pure joints, sheared joints, and faults, ranging in length from 0.25 to 5,444 m and total maximum displacement from 0.005 to 87.95 m (Figure 16, Table 1). We include a total of 75 fractures, including 53 pure joints, 18 sheared joints, the Ohale and White Rabbit Faults, and the two minor faults. The Kulanaokuaiki Fault was excluded because the collapsed monocline rendered measurements of the total displacement along the structure impossible. The four measurements of faults with monoclines have substantially longer lengths and larger maximum total displacements, clustering in a separate region of the diagram, while pure and sheared joints are clustered together (Figure 16). Our measurements of pure joints have a broader length range, spanning from 0.25 to 200 m, whereas sheared joints have a more limited length range, ranging from 1.5 to 61 m.

The fit to the data of all fracture types yields a power-law relationship with an exponent of 0.88 and $R^2 = 0.92$ (Figure 16), showing a good correlation of total maximum displacement with length. This scaling is sublinear, that is, having an exponent < 1 , falling in between the slope of 1 expected for fault scaling and the slope of 0.5 for opening-mode fractures. There is no break in slope between our joint and fault measurements, indicating that joint apertures also have a nearly linear scaling relationship with their lengths, not following the square root relationship found typical for opening-mode fractures (Figure 1b).

To explore the scaling of the different displacement components, we decompose the maximum total displacement vectors (Figure 15) and plot them individually (Figure 17). This allows us to investigate the scaling of apertures of the pure joints (Figure 17a), openings ($A + S_h$) of sheared joints (Figure 17b), and the structural relief (S_v) of both sheared joints and faults with monoclines (Figure 17c). Since it is unknown to what extent the width of the monoclines contributes to the horizontal displacement, our width measurements for faults with monoclines are excluded from this analysis.

The maximum apertures ($A_{\max}$) of pure joints range from 0.005 to 3.08 m. They show a moderate to low correlation with length, whereby the best-fit power-law function has an exponent of 0.858 ($R^2 = 0.47$, Figure 17a). Power-law exponents below unity indicate a sublinear scaling as is found typical for opening-mode fractures, but the value is higher than the theoretical exponent of 0.5 (Olson, 2003). The maximum openings of sheared joints ($A + S_h$) range from 0.03 to 5.3 m. The best-fit power-law function indicates that the openings of sheared joints scale with lengths, showing an exponent of 0.67, but with very poor to no correlation ($R^2 = 0.06$, Figure 17b). The

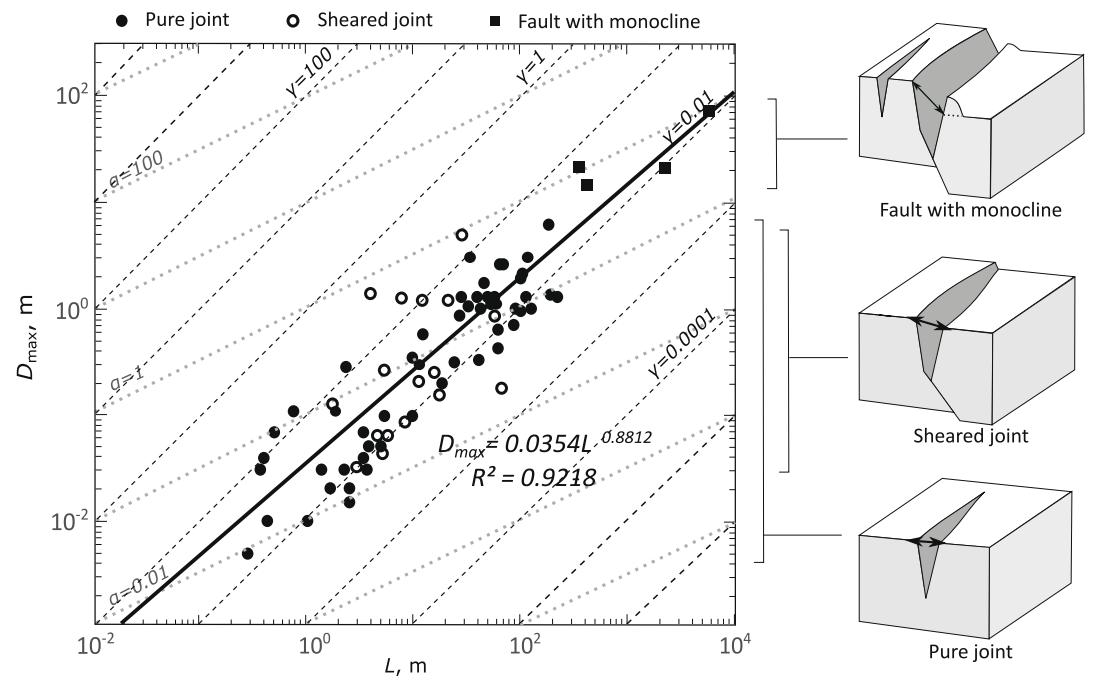


Figure 16. Displacement-to-length scaling plot of all measured fractures along the Koa'e Fault Zone, showing maximum total displacements ($D_{\max}$) as a function of their lengths (L). The data for pure joints are plotted as black solid dots, sheared joints as black open circles, and faults with monoclines as black squares. The block diagrams on the right show the ranges of $D_{\max}$ for each fracture type. For reference, the expected slopes for opening-mode and sliding/tearing-mode fractures are plotted in gray dotted and black dashed lines, respectively.

maximum structural reliefs (S_v)_{max} of sheared joints and faults with monoclines range from 0.01 to 0.84 m and from 8.3 to 81 m, respectively (Figure 17c). As the structural relief represents the fault throw, and because the fault dip is unknown, it is the best approximation for the fault slip displacement component of these mixed-mode fractures. The reliefs show a moderate correlation with length, following the best-fit power-law function with an exponent of 0.89 ($R^2 = 0.65$, Figure 17c). This exponent is close to the theoretical exponent of 1 for fault scaling (see Figure 1a).

The scaling of opening-mode fractures with an exponent around 0.5 is evident for pure joints from literature data (Figure 1b; e.g., Olson, 2003; Schultz et al., 2013), implying that the propagation of these fractures is governed by the fracture toughness, which in turn influences their length and maximum apertures. The relationship of the maximum apertures and lengths of the pure joints along the Koa'e Fault Zone, however, is not well described by sublinear scaling but is closer to linear scaling (Figure 17a). This may be because all of these pure joints utilize pre-existing columnar joints of the basalt, and thus, the propagation may no longer be fully controlled by the fracture toughness. The increase in the exponent value can also result from post-propagation conditions, where the joints undergo relaxation and change aperture without growth in fracture lengths (e.g., Olson, 2003). The pure joints in this study also have various degrees of interactions and linkage with other fractures (such as in Figures 13a and 13c), which typically have a higher scaling exponent with a large scatter (e.g., Hatton et al., 1993).

The linear scaling of all fractures in the Koa'e Fault Zone with an exponent of ~ 1 (Figure 11), is typical for faulting, well documented in numerous literature (see Figure 1a). This scaling implies these fractures propagate with constant shear driving stress, controlled by the yield strength of the rock (e.g., Cowie & Scholz, 1992), which may also apply to the fracture growth of the Koa'e Fault Zone. The exponent of 0.88 for all fractures being less than 1 could be attributed to pure jointing occurring in the earlier stages of the mixed-mode fracture development, where the exponent from the maximum aperture-to-length scaling of pure joints (Figure 17a) is higher than 0.5 but still below 1, as previously explained.

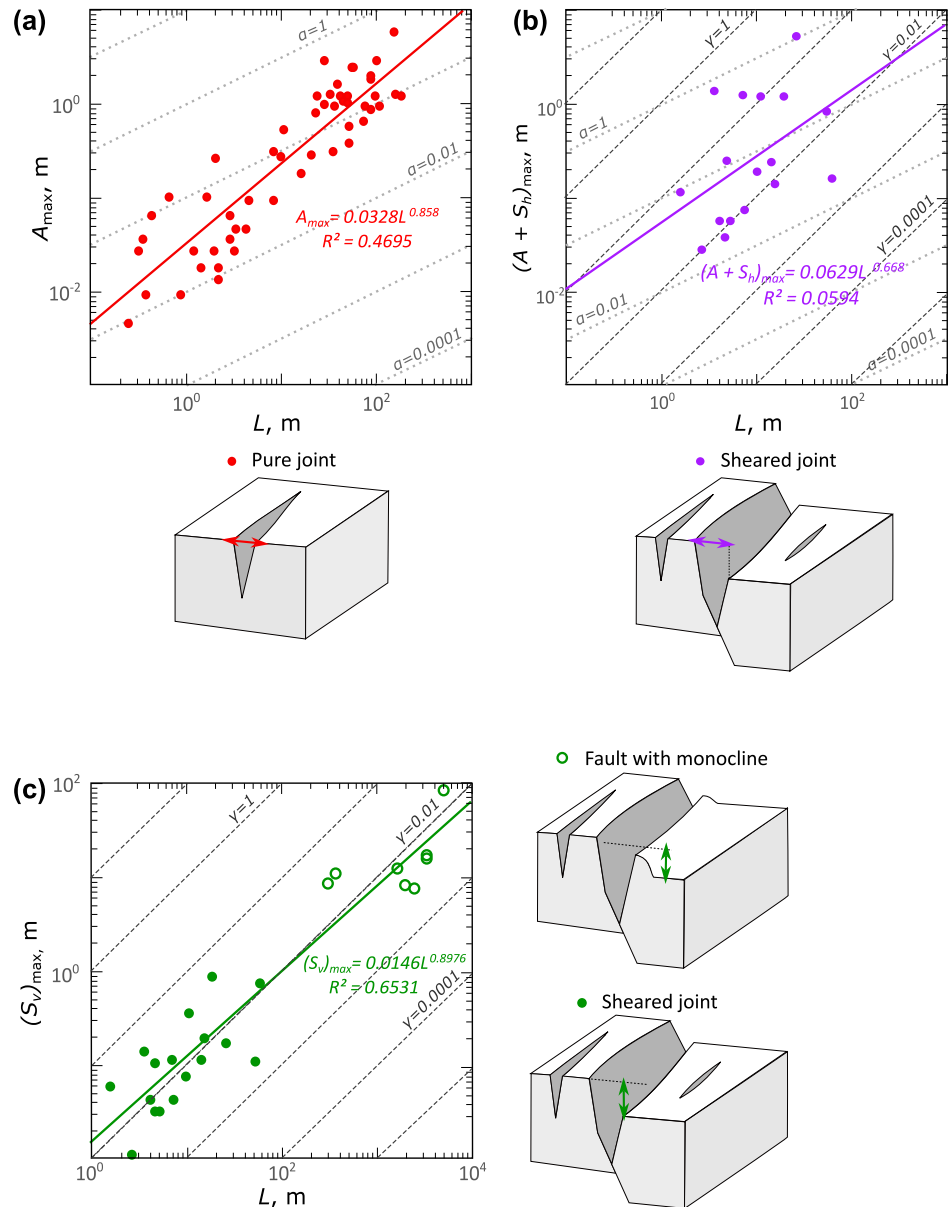


Figure 17. Displacement-to-length scaling plots of the vector components of maximum displacements of all measured fractures along the Koa'e Fault Zone. (a) Displacement-to-length scaling of pure joints showing the maximum apertures ($A_{\max}$) as a function of joint length. (b) Displacement-to-length scaling of sheared joints showing their maximum openings ($(A + S_h)_{\max}$) as a function of length. The plot does not include the measurement for faults with monoclines. (c) Displacement-length scaling of sheared joints and main faults with monoclines showing the maximum relief ($(S_v)_{\max}$) as a function of length. The theoretical slopes for opening-mode and sliding/tearing mode fractures are plotted in gray dotted lines and black dashed lines, respectively.

Intrinsic scatter is well recognized in displacement-to-length scaling data due to fracture orientation, joint network geometry, mechanical interactions, three-dimensional shape, and rock mass properties (see Klimczak et al., 2010). Given the dips of the fault planes along the Koa'e Fault Zone are not well-constrained, their variation in different fracture planes may also contribute to the scatter in the scaling (Figures 16 and 17). The estimated fault dips along the Kulanaokuaiki Fault range from 65° to 78° derived from the focal mechanisms of a sequence of three earthquakes in 2012 (Ge et al., 2019). However, it is unclear if these dips are representative of all structures in this population. Since the total mixed-mode displacement is independent of the fault dip (i.e., the

opening is always perpendicular to the relief; Figure 15), it may account for the smaller scatter in its scaling with length, resulting in the high R^2 value of 0.92 (Figure 16).

In combination with the observed structural evolution (Figure 14), we interpret the $D_{\max}/L$ scaling behavior of the Koa'e Fault Zone to capture the transition from jointing to faulting. Such fracture scaling has heretofore largely remained unexplored, prompting further discussion on mixed-mode fracture scaling involving opening and sliding/tearing modes. Whether joints transition to faults or joints constituting the dominant fracture type in a fault damage zone, both joints and faults occurring as part of the same fracture population is important for understanding fluid migration along these structures. At present, however, it is unclear how this scaling applies in detail to other mixed-mode fracture systems. Knowledge of the scaling behavior of more such fracture populations may allow for better quantification of flow rates or storage volumes of faulted crystalline rock (e.g., Klimczak et al., 2010).

8. Conclusion

To investigate the scaling of mixed-mode fractures showing opening-, and sliding/tearing modes, we conducted structural mapping along the Koa'e Fault Zone. We collected structural observations and measurements of fracture displacements, including their relief and opening components, in regular intervals along several faults and joints of various sizes. We combined our geological observations to develop a structural model of the fault zone evolution (Figure 9). Subvertical jointing is initiated at the surface when tensile stresses from the rifting of the area reach the strength of the rock mass. When the joints propagate downwards, they transition to faulting at a depth where the appropriate loading conditions are reached. The joints are sheared with relief and dilation added as additional displacement components to the original aperture. With the increasing growth of the structures, they develop into normal faults accompanied by a monoclinical flexure in the hanging wall. When faults continue to grow along non-vertically dipping fault planes, a dilational displacement component that results from the sliding of the fault continues to widen the openings formed previously by the jointing. Pure joints, sheared joints, and minor faults are found not only in association with the main faults but also in isolated locations away from the main faults, where none of these sheared joints have a monocline. Furthermore, en echelon joints are commonly observed as process zones at fault tips. These observations support a fault zone development from downward propagation of joints that transition to faulting at depth.

We established a displacement-to-length scaling relationship of the Koa'e Fault Zone that captures the transition from pure jointing to faulting. Our scaling reveals that the scaling of the entire mixed-mode fracture system has a power-law scaling exponent of 0.88, similar to the typical scaling of faults rather than joints, with no break in slope from jointing to faulting. The maximum aperture-to-length scaling of pure joints shows a larger exponent value than the expected theoretic value of 0.5 ascribe to a series of reasons, including the influence of pre-existing joints, post-propagation conditions, and fracture interactions and linkages. Moreover, the scatter in scaling data may result from different factors, including the unknown fracture dips and their variation in different fracture planes along the fault zone. The results presented in this paper are important for resolving the development and scaling behavior of this type of mixed-mode fractures, allowing for better quantification of flow rates or fluid storage volumes of faulted crystalline rock.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Supplemental Material includes the displacement profiles of sheared joints (Figures S1 in Supporting Information S1) with a description of their structural morphologies, all field data with GPS record (Table S1 in Supporting Information S2), organized data set for the displacement profiles (Table S2 in Supporting Information S3), compiled displacement-length scaling data set (Table S3 in Supporting Information S4), and the shapefile of the GPS record (Shapefile S10 in Supporting Information S5). The data files used in this paper are available at (Cheng, 2024).

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